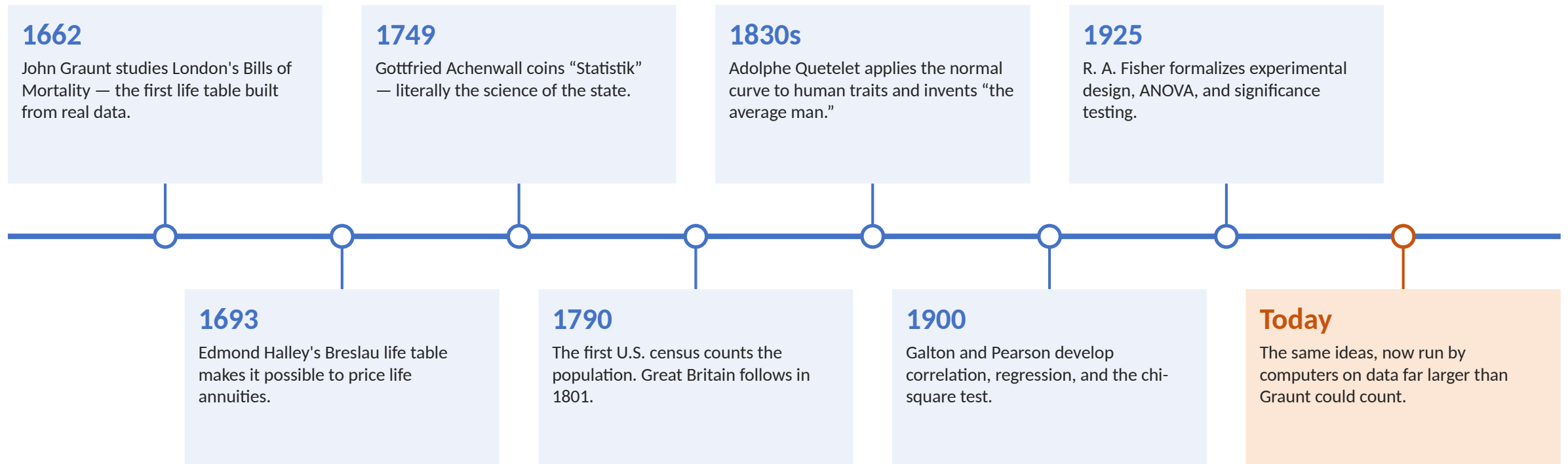


Lecture 1

Introduction to Statistical Methods

Where does statistics come from?

Statistics began as bookkeeping about people — counting births, deaths, and taxpayers so that a state could describe itself.



Sources: J. Graunt (1662), Natural and Political Observations Made upon the Bills of Mortality. E. Halley (1693), Phil. Trans. R. Soc. 17:596–610. G. Achenwall (1749), Abriss der neuesten Staatswissenschaft. A. Quetelet (1835), Sur l'homme et le développement de ses facultés. R. A. Fisher (1925), Statistical Methods for Research Workers.

Answering Questions with data

- What proportion of people in the U.S are biologically female?
Survey Americans and ask them to report their sex
- Does a low-carb diet result in significant weight loss?
 - Design an experiment to evaluate the effectiveness of the low carb diet on weight loss
 - Record information such as starting and ending weight, calories consumed per day, ...
- Are people more likely to stop at Starbucks if they've recently seen a Starbucks TV add?
 - Conduct a marketing survey to record the number of people who have gone to starbucks since add aired
 - Compare between those who saw the add and those who did not



Where do we find Statistics? – Everywhere!

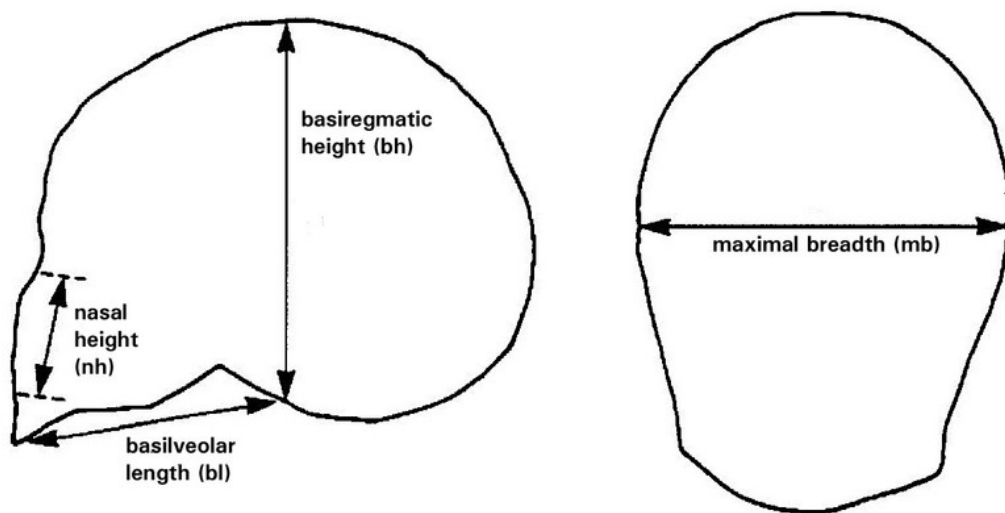
- **Sports** (number of points scored per game by an athlete)
- **Weather** Forecasting (average monthly rainfall)
- **Economy** (median income, unemployment rate)
- **Sales Tracking** (projected sales revenue from iphone 14's sold in a given month)
- **Medicine** (percentage of people who were pain free from using a drug)
- **Manufacturing** (number of cans of coke produced in a given month)

The **science of statistics** deals with the collection, analysis, interpretation, and presentation of data

Data is a collection of observations/measurements

What Do Data Look Like?

The following are data from an analysis of 150 Egyptian skulls from 5 epochs of Egyptian history. For each skull, the epoch, and several measurements characterizing the shape of the skull are recorded



Source: D. J. Hand, F. Daly, A. D. Lunn, K. J. McConway and E. Ostrowski (1994). A Handbook of Small Datasets, Chapman and Hall/CRC, London.

Visualizing Tests for Equality of Covariance Matrices - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Diagram-of-the-skull-measurements-for-the-Egyptian-skulls-data-set-Maximal-breadth-and_fig4_325126938 [accessed 21 Aug, 2023]

Observation	epoch	mb	bh	bl	nh
1	c4000BC	131	138	89	49
2	c4000BC	125	131	92	48
3	c4000BC	131	132	99	50
4	c4000BC	119	132	96	44
5	c4000BC	136	143	100	54
6	c4000BC	138	137	89	56
7	c4000BC	139	130	108	48
8	c4000BC	125	136	93	48
9	c4000BC	131	134	102	51
10	c4000BC	134	134	99	51
11	c4000BC	129	138	95	50
12	c4000BC	134	121	95	53
13	c4000BC	126	129	109	51
14	c4000BC	132	136	100	50
15	c4000BC	141	140	100	51
⋮	⋮	⋮	⋮	⋮	⋮
145	cAD150	132	127	97	52
146	cAD150	137	125	85	57
147	cAD150	129	128	81	52
148	cAD150	140	135	103	48
149	cAD150	147	129	87	48
150	cAD150	136	133	97	51

Example Data 2

The following table contains data on 20 clients contacted during a bank’s direct marketing campaign for a term deposit account. For each client, the job, education level, age, account balance, length of the marketing call, and whether they subscribed are recorded

Reading a data table

Each column is a variable

Client ID	Job	Age	Balance	Subscribed
C-1047	mgmt	42	2,143	no
C-1052	blue	35	413	no
C-1068	tech	29	876	no
C-1073	admin	47	1,584	no

Each row is one observation — one client

Observation	Client ID	Job	Education	Age	Balance (\$)	Call Length (min)	Subscribed
1	C-1047	management	tertiary	42	2,143.50	4.6	no
2	C-1052	blue-collar	primary	35	412.75	2.1	no
3	C-1068	technician	secondary	29	876.20	5.8	no
4	C-1073	admin.	secondary	47	1,584.05	3.2	no
5	C-1089	retired	primary	68	3,921.40	9.7	yes
6	C-1094	services	secondary	31	-88.60	1.4	no
7	C-1101	management	tertiary	38	4,507.85	11.2	yes
8	C-1117	blue-collar	secondary	52	233.15	2.8	no
9	C-1126	self-employed	tertiary	44	1,902.30	6.5	no
10	C-1138	technician	secondary	26	145.90	3.9	no
11	C-1145	admin.	tertiary	33	2,760.65	8.1	yes
12	C-1159	entrepreneur	tertiary	50	6,318.20	4.3	no
13	C-1163	student	secondary	22	318.45	7.4	no
14	C-1178	blue-collar	primary	58	709.10	1.9	no
15	C-1184	services	secondary	40	1,067.35	5.1	no
16	C-1190	management	tertiary	45	3,245.75	2.6	no
17	C-1203	retired	secondary	71	2,884.00	10.5	yes
18	C-1215	technician	tertiary	36	952.60	4.8	no
19	C-1228	admin.	secondary	49	1,471.25	3.5	no
20	C-1234	blue-collar	secondary	27	-45.30	6.2	no

Source: Variable structure adapted from Moro, S., Rita, P., and Cortez, P. (2014). Bank Marketing [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5K306>. The 20 client records shown here are simulated for illustration; the real data set contains 45,211 clients.

Checkpoint:



Data - a collection of observations/measurements on a set of variables - typically represented as a table.

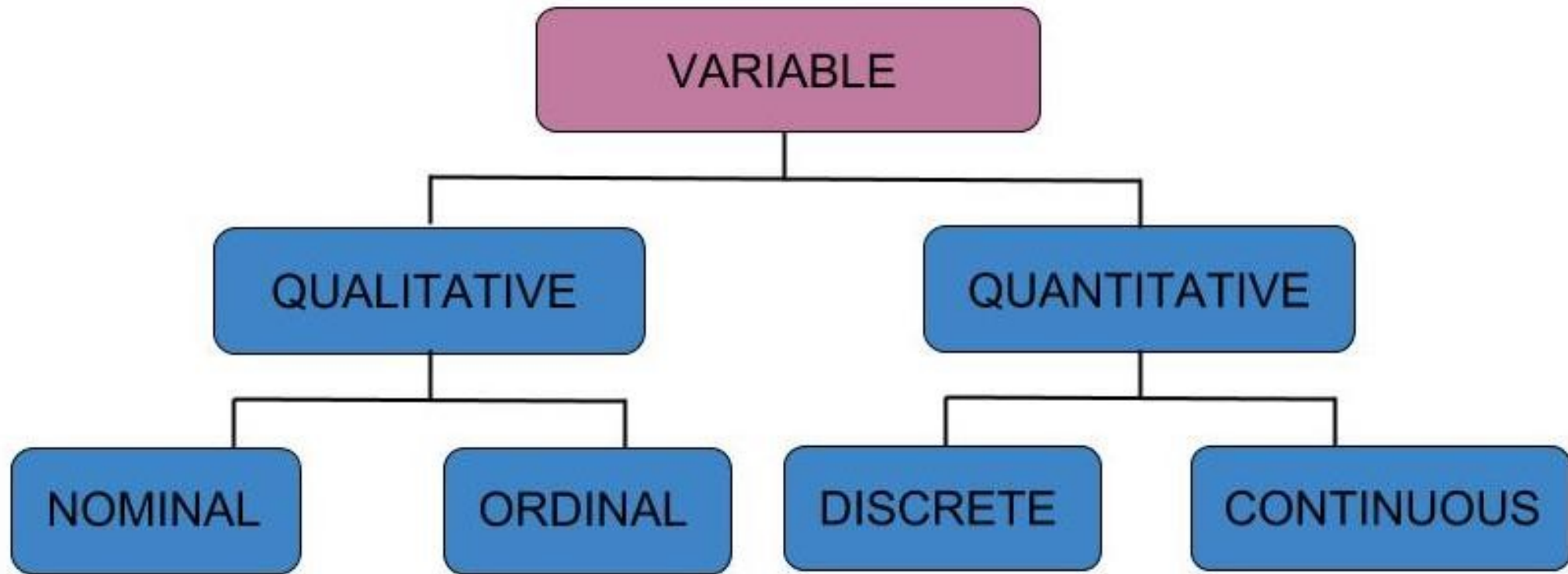


Observation – the fundamental unit of data – typically observations are rows of a data table



Variables – characteristics of an observation – typically variables are the columns of a data table

Types of Variables: Qualitative vs Quantitative



Types of Variables: Qualitative vs Quantitative

Qualitative (Categorical) variable – non-numeric qualities or characteristics that can be placed in distinct categories

State/city (ID, WA, MT, ...)

Treatment (Drug/Placebo)

Genotype (AA, AT, TT)

Survival (live or die)



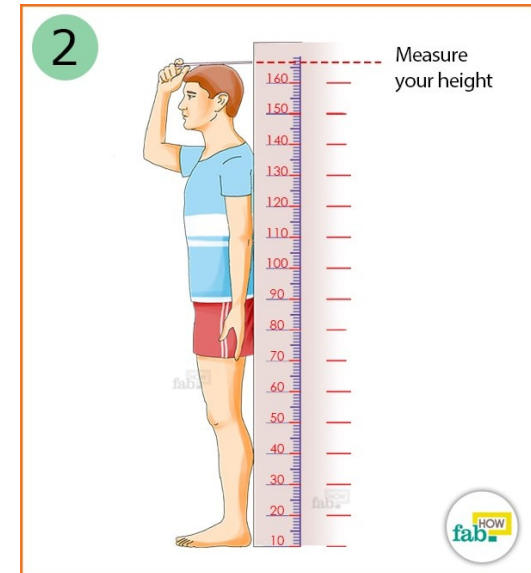
Quantitative (numeric) variable – numerical characteristics - can be ordered or ranked

Height (inches/cm)

Weight (lbs/kg)

Longevity/Age (number of years)

Dose (micrograms per gram)



Practice: Qualitative vs Quantitative

From our example data of egyptian skulls, which variables are qualitative and which are quantitative?

Qualitative: Epoch

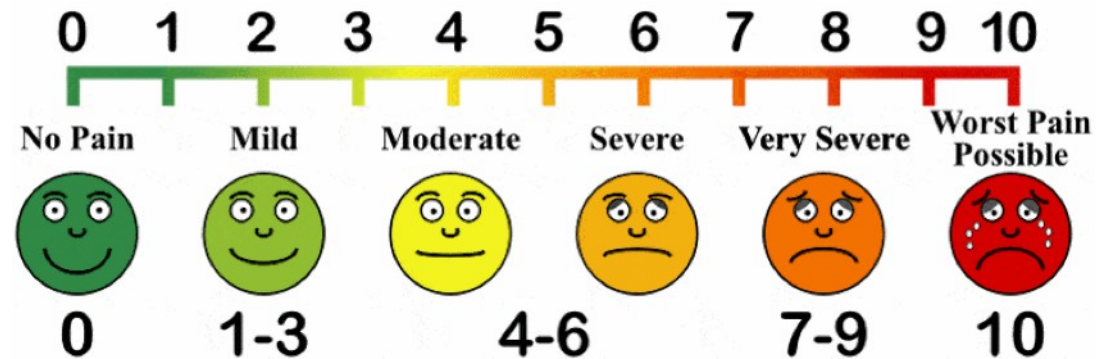
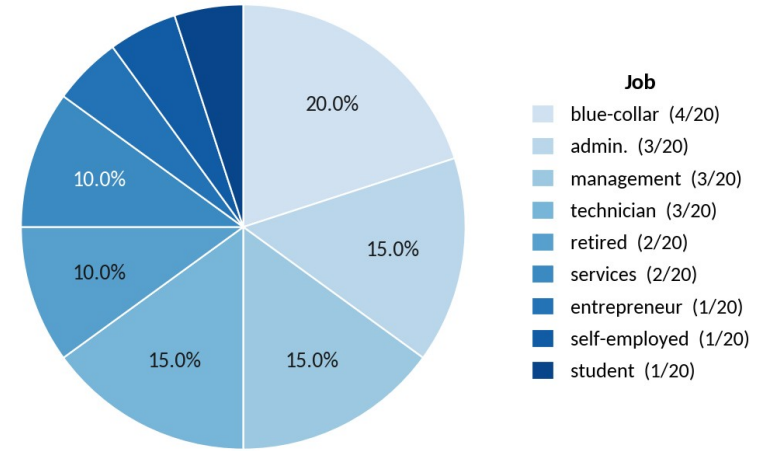
Quantitative: mb, bh, bl, nh

Observation	epoch	mb	bh	bl	nh
1	c4000BC	131	138	89	49
2	c4000BC	125	131	92	48
3	c4000BC	131	132	99	50
4	c4000BC	119	132	96	44
5	c4000BC	136	143	100	54
6	c4000BC	138	137	89	56
7	c4000BC	139	130	108	48
8	c4000BC	125	136	93	48
9	c4000BC	131	134	102	51
10	c4000BC	134	134	99	51
11	c4000BC	129	138	95	50
12	c4000BC	134	121	95	53
13	c4000BC	126	129	109	51
14	c4000BC	132	136	100	50
15	c4000BC	141	140	100	51
⋮	⋮	⋮	⋮	⋮	⋮
145	cAD150	132	127	97	52
146	cAD150	137	125	85	57
147	cAD150	129	128	81	52
148	cAD150	140	135	103	48
149	cAD150	147	129	87	48
150	cAD150	136	133	97	51

Types of Variables: Qualitative vs Quantitative

Qualitative nominal – non-numeric qualities or characteristics that can be placed in distinct categories that do not have a natural ordering (e.g labels, names, colors, etc)

Qualitative ordinal – non-numeric qualities or characteristics that can be placed in distinct categories with an inherent ordering (Likert responses, education level, military ranking)



This Photo by Unknown Author is licensed under [CC BY-NC-ND](https://creativecommons.org/licenses/by-nc-nd/4.0/)

Practice: Qualitative vs Quantitative

In the bank marketing example data, which variable(s) are qualitative nominal and which are qualitative ordinal?

Qualitative nominal: Client ID, Job, Subscribed

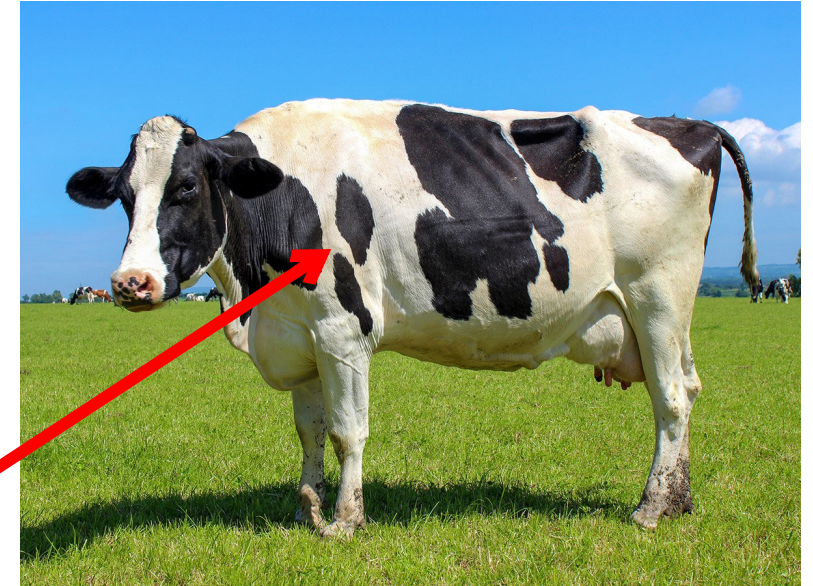
Qualitative ordinal: Education

Observation	Client ID	Job	Education	Age	Balance (\$)	Call Length (min)	Subscribed
1	C-1047	management	tertiary	42	2,143.50	4.6	no
2	C-1052	blue-collar	primary	35	412.75	2.1	no
3	C-1068	technician	secondary	29	876.20	5.8	no
4	C-1073	admin.	secondary	47	1,584.05	3.2	no
5	C-1089	retired	primary	68	3,921.40	9.7	yes
6	C-1094	services	secondary	31	-88.60	1.4	no
7	C-1101	management	tertiary	38	4,507.85	11.2	yes
8	C-1117	blue-collar	secondary	52	233.15	2.8	no
9	C-1126	self-employed	tertiary	44	1,902.30	6.5	no
10	C-1138	technician	secondary	26	145.90	3.9	no
11	C-1145	admin.	tertiary	33	2,760.65	8.1	yes
12	C-1159	entrepreneur	tertiary	50	6,318.20	4.3	no
13	C-1163	student	secondary	22	318.45	7.4	no
14	C-1178	blue-collar	primary	58	709.10	1.9	no
15	C-1184	services	secondary	40	1,067.35	5.1	no
16	C-1190	management	tertiary	45	3,245.75	2.6	no
17	C-1203	retired	secondary	71	2,884.00	10.5	yes
18	C-1215	technician	tertiary	36	952.60	4.8	no
19	C-1228	admin.	secondary	49	1,471.25	3.5	no
20	C-1234	blue-collar	secondary	27	-45.30	6.2	no

Quantitative Variables: Discrete vs Continuous

- **Quantitative discrete** – quantitative variables that take on distinct, countable values such 0,1,2,3... (whole numbers or integers)
 - all counts are quantitative discrete variables

e.g. number of black spots on a dairy cow



- **Quantitative continuous** – variables that can take on infinite number of values within an interval of any two specific values (e.g. temperature °F, height in inches, speed in miles per hour)



Practice: Discrete vs Continuous

In the bank marketing example data, which variable(s) are quantitative discrete and which are quantitative continuous?

Quantitative discrete: Age (a count of the number of years)

Quantitative continuous: Balance, Call Length

Observation	Client ID	Job	Education	Age	Balance (\$)	Call Length (min)	Subscribed
1	C-1047	management	tertiary	42	2,143.50	4.6	no
2	C-1052	blue-collar	primary	35	412.75	2.1	no
3	C-1068	technician	secondary	29	876.20	5.8	no
4	C-1073	admin.	secondary	47	1,584.05	3.2	no
5	C-1089	retired	primary	68	3,921.40	9.7	yes
6	C-1094	services	secondary	31	-88.60	1.4	no
7	C-1101	management	tertiary	38	4,507.85	11.2	yes
8	C-1117	blue-collar	secondary	52	233.15	2.8	no
9	C-1126	self-employed	tertiary	44	1,902.30	6.5	no
10	C-1138	technician	secondary	26	145.90	3.9	no
11	C-1145	admin.	tertiary	33	2,760.65	8.1	yes
12	C-1159	entrepreneur	tertiary	50	6,318.20	4.3	no
13	C-1163	student	secondary	22	318.45	7.4	no
14	C-1178	blue-collar	primary	58	709.10	1.9	no
15	C-1184	services	secondary	40	1,067.35	5.1	no
16	C-1190	management	tertiary	45	3,245.75	2.6	no
17	C-1203	retired	secondary	71	2,884.00	10.5	yes
18	C-1215	technician	tertiary	36	952.60	4.8	no
19	C-1228	admin.	secondary	49	1,471.25	3.5	no
20	C-1234	blue-collar	secondary	27	-45.30	6.2	no